

| Parameter                      | Value/Setting                                       |
|--------------------------------|-----------------------------------------------------|
| Simulation id                  | 7ea5f1a1                                            |
| Experiment                     | Collocated                                          |
| Git commit                     | 78bd749                                             |
| Wall time (s)                  | 25582.99                                            |
| Reynolds number                | 20                                                  |
| Mesh type                      | Unstructured                                        |
| Number of control volumes      | 87542                                               |
| Boundary conditions            | shared_configs/domain/boundaries__cylinderFlow.yaml |
| Algorithm                      | SIMPLE                                              |
| Convection scheme              | TVD                                                 |
| Limiter                        | MUSCL                                               |
| Convergence tolerance          | 1e-7                                                |
| Momentum relaxation            | 0.5                                                 |
| Pressure relaxation            | 0.1                                                 |
| Momentum solver                | bcgs                                                |
| Momentum solver preconditioner | hypre                                               |
| Momentum solver tolerance      | 1e-8                                                |
| Pressure solver                | bcgs                                                |
| Pressure solver preconditioner | hypre                                               |
| Pressure solver tolerance      | 1e-10                                               |
| Non-orthogonal corrections     | 1                                                   |
| Number of iterations           | 20000                                               |
| Final u-residual               | 1.49e-6                                             |
| Final v-residual               | 4.03e-7                                             |
| Final continuity-residual      | 1.58e-4                                             |
| results.drag_coefficient       | 0.6410312973686818                                  |
| results.lift_coefficient       | 0.016477361862676187                                |
| results.pressure_difference    | 0.2516809464784178                                  |